Notes for PHYS 34202 - Introduction To Quantum Science

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Contents

Course Description

This course offers an introduction to quantum mechanics, exploring how it has transformed our understanding of the physical world and revolutionized modern technology. The concepts will be illustrated with a variety of contemporary physics results and applications, including relativistic effects and a brief overview of the emerging field of quantum information science and engineering.

Particle-Wave Duality and Superposition

> Newton thought that light was made up of particles, but then it was discovered, as we have seen here, that it behaves like a wave. Later, however (in the beginning of the twentieth century) it was found that light did indeed sometimes behave like a particle. Historically, the electron, for example, was thought to behave like a particle, and then it was found that in many respects it behaved like a wave. So it really behaves like neither. Now we have given up. We say: “It is like neither."
>
> *Richard Feynman*

Quantum means quantized, as in, coming in integers. You may have one atom, or a million, but you may not have half an atom (unless you are really unlucky!). Waves may also be quantized. Think of two children spinning a jumprope; the rope waves regularly, in one place. The children may change the shape of the wave(s), the period, and other properties.

Whenever you measure a system, you disturb it. In classical physics this disturbance does not interfere with the system much. A falling ball or pendulum is unbothered by photons bouncing off and entering the eyes of physicists. However, a particle only slightly bigger than a photon may be very disturbed when we bounce a laser off of it to see what it’s doing! When we measure a quantum particle, it forces the particle into an allowed state. The particle will remain in that state as long as it’s not disturbed, and we can keep measuring if it’s in that state.